

CBSE Class 10 Mathematics — Areas Related to Circles — Solutions

1. 1. If the circumference of a circle is 176 cm, then its radius is:

Answer: (C)

Solution:

1. Use the formula for circumference $C = 2\pi r$.
2. Substitute $C = 176$ and $\pi = \frac{22}{7}$.
3. Solve for r .

Derivation:

$$\begin{aligned}176 &= 2 \times \frac{22}{7} \times r \\ \implies 176 &= \frac{44}{7} \times r \\ \implies r &= \frac{176 \times 7}{44} = 4 \times 7 = 28 \text{ cm}\end{aligned}$$

Areas Related to Circles — Circumference of a Circle

2. 2. The area of a circle whose diameter is 14 cm is:

Answer: (B)

Solution:

1. Find the radius $r = \frac{d}{2} = \frac{14}{2} = 7$ cm.
2. Use the area formula $A = \pi r^2$.
3. Substitute $r = 7$ and compute.

Derivation:

$$A = \frac{22}{7} \times 7^2 = \frac{22}{7} \times 49 = 22 \times 7 = 154 \text{ cm}^2$$

Areas Related to Circles — Area of a Circle

3. 3. The area of a quadrant of a circle with radius r is:

Answer: (D)

Solution:

1. A quadrant is one-fourth of a circle.
2. Area of circle $= \pi r^2$.
3. Area of quadrant $= \frac{1}{4}\pi r^2$.

Derivation:

$$\text{Area} = \frac{90^\circ}{360^\circ} \times \pi r^2 = \frac{1}{4}\pi r^2$$

Areas Related to Circles — Area of a Sector

4. **4.** If the perimeter of a semi-circular protractor is 36 cm, its diameter is:

Answer: (A)

Solution:

1. Perimeter of a semi-circle is $\pi r + 2r = r(\pi + 2)$.
2. Set $r(\frac{22}{7} + 2) = 36$.
3. Solve for r and then diameter $d = 2r$.

Derivation:

$$\begin{aligned}r\left(\frac{22 + 14}{7}\right) &= 36 \\ \implies r\left(\frac{36}{7}\right) &= 36 \\ \implies r &= 7. \text{ Thus, } d = 2 \times 7 = 14 \text{ cm.}\end{aligned}$$

Areas Related to Circles — Perimeter of Semi-circle

5. **5.** The area of a sector of a circle with radius 6 cm and central angle 60° is:

Answer: (C)

Solution:

1. Use formula $A = \frac{\theta}{360} \times \pi r^2$.
2. Substitute $\theta = 60$, $r = 6$.

Derivation:

$$A = \frac{60}{360} \times \pi \times 6^2 = \frac{1}{6} \times 36\pi = 6\pi \text{ cm}^2$$

Areas Related to Circles — Area of a Sector

6. **6.** Assertion (A): The area of a circle with radius 7 cm is 154 cm^2 . Reason (R) : The area of a circle is given by the formula πr^2 .

Answer: (A)

Solution:

1. Calculate area using πr^2
2. Area = $(22/7) \times 7 \times 7 = 154 \text{ cm}^2$
2. Both statements are true and R is the correct explanation

Derivation:

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 7^2 = 154$$

Areas Related to Circles — Area of a Circle

7. **7.** Assertion (A): The circumference of a circle is 44 cm, then its radius is 7 cm. Reason (R): Circumference of a circle is $2\pi r$.

Answer: (A)

Solution:

1. Use $2\pi r = 44$

$$2. 2 \times (22/7) \times r = 44$$

$$3. r = (44 \times 7)/44 = 7 \text{ cm}$$

Derivation:

$$2 \times \frac{22}{7} \times r = 44$$

$$\implies r = 7$$

Areas Related to Circles — Circumference of a Circle

8. **8.** Assertion (A): The area of a semicircle of radius r is $\frac{1}{2}\pi r^2$. Reason (R): A semicircle is half of a circle.

Answer: (A)

Solution:

1. Area of circle is πr^2
2. Semicircle is exactly half, so area is $(1/2)\pi r^2$
3. Both statements are true and R is the correct explanation

Derivation:

$$Area_{semicircle} = \frac{1}{2} \times Area_{circle} = \frac{1}{2}\pi r^2$$

Areas Related to Circles — Area of Semicircle

9. **9.** Assertion (A): If the perimeter of a circle is equal to that of a square, then the area of the circle is greater than the area of the square. Reason (R): The area of a circle with a given perimeter is always the maximum possible area for any plane figure.

Answer: (A)

Solution:

1. Let perimeter be P . Circle $r = P/2$, Area = $P^2/4$. Square side = $P/4$, Area = $P^2/16$.
2. Since $4\pi \approx 12.56 < 16$, $P^2/12.56 > P^2/16$.
3. Both statements are correct.

Derivation:

$$A_{circle} = \frac{P^2}{4\pi} > A_{square} = \frac{P^2}{16}$$

Areas Related to Circles — Comparison of Areas

10. **10.** Assertion (A): The length of an arc of a sector of a circle with radius r and angle θ is $\frac{\theta}{360} \times 2\pi r$. Reason (R): The circumference of a circle is $2\pi r$.

Answer: (A)

Solution:

1. An arc is a fraction of the circumference determined by the angle θ .
2. The fraction is $\theta/360$. Multiply by total circumference $2\pi r$.
3. Both are true and R explains the basis of the formula.

Derivation:

$$ArcLength = \frac{\theta}{360} \times 2\pi r$$

Areas Related to Circles — Length of an Arc

- 11. 11.** Find the area of a sector of a circle with radius 6 cm if the angle of the sector is 60° . (Use $\pi = \frac{22}{7}$)

Answer:

$$\frac{132}{7} \text{ cm}^2$$

Solution:

1. Identify the formula for the area of a sector as $\frac{\theta}{360^\circ} \times \pi r^2$.
2. Substitute $r = 6$ and $\theta = 60^\circ$ into the formula and simplify.

Derivation:

$$Area = \frac{60}{360} \times \frac{22}{7} \times 6^2 = \frac{1}{6} \times \frac{22}{7} \times 36 = 6 \times \frac{22}{7} = \frac{132}{7} \text{ cm}^2$$

Areas Related to Circles — Area of Sector

- 12. 12.** The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

Answer:

$$\frac{154}{3} \text{ cm}^2$$

Solution:

1. Calculate the angle swept by the minute hand in 1 minute ($360^\circ/60 = 6^\circ$), so for 5 minutes, $\theta = 30^\circ$.
2. Apply the area of sector formula with $r = 14$ and $\theta = 30^\circ$.

Derivation:

$$Area = \frac{30}{360} \times \frac{22}{7} \times 14^2 = \frac{1}{12} \times \frac{22}{7} \times 196 = \frac{1}{12} \times 22 \times 28 = \frac{154}{3} \text{ cm}^2$$

Areas Related to Circles — Application of Sector Area

- 13. 13.** Find the area of a quadrant of a circle whose circumference is 44 cm. (Use $\pi = \frac{22}{7}$)

Answer:

$$38.5 \text{ cm}^2$$

Solution:

1. Use the circumference formula $2\pi r = 44$ to find the radius r .
2. Calculate the area of the quadrant using $\frac{1}{4}\pi r^2$.

Derivation:

$$2 \times \frac{22}{7} \times r = 44$$
$$\Rightarrow r = 7 \text{ cm. Area} = \frac{1}{4} \times \frac{22}{7} \times 7^2 = \frac{1}{4} \times 22 \times 7 = 38.5 \text{ cm}^2$$

Areas Related to Circles — Area of Circle Parts

- 14. 14.** If the sum of the areas of two circles with radii R_1 and R_2 is equal to the area of a circle of radius R , prove that $R_1^2 + R_2^2 = R^2$.

Answer:

$$R_1^2 + R_2^2 = R^2$$

Solution:

1. Write the equation based on the given condition: $\pi R_1^2 + \pi R_2^2 = \pi R^2$.
2. Divide the entire equation by π to obtain the result.

Derivation:

$$\pi R_1^2 + \pi R_2^2 = \pi R^2$$
$$\Rightarrow \pi(R_1^2 + R_2^2) = \pi R^2$$
$$\Rightarrow R_1^2 + R_2^2 = R^2$$

Areas Related to Circles — Area of Circle

- 15. 15.** A chord of a circle of radius 10 cm subtends a right angle at the center. Find the area of the corresponding minor segment. (Use $\pi = 3.14$)

Answer:

$$28.5 \text{ cm}^2$$

Solution:

1. Calculate the area of the sector: $\frac{90}{360} \times \pi r^2$.
2. Subtract the area of the triangle $\frac{1}{2}r^2 \sin(90^\circ)$ from the sector area.

Derivation:

$$\text{Area} = \frac{90}{360} \times 3.14 \times 100 - \frac{1}{2} \times 10 \times 10 = 78.5 - 50 = 28.5 \text{ cm}^2$$

Areas Related to Circles — Area of Segment

- 16. 16.** Find the area of the shaded region in a circle of radius 14 cm, where a sector subtends an angle of 60° at the center. (Use $\pi = \frac{22}{7}$)

Answer:

$$102.67 \text{ cm}^2$$

Solution:

1. Identify the formula for the area of a sector: $A = \frac{\theta}{360^\circ} \times \pi r^2$.
2. Substitute the given values: $\theta = 60^\circ$ and $r = 14 \text{ cm}$.

3. Calculate the area: $\frac{60}{360} \times \frac{22}{7} \times 14 \times 14$.

Derivation:

$$Area = \frac{60}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{1}{6} \times 22 \times 2 \times 14 = \frac{616}{6} = 102.67 \text{ cm}^2$$

Areas Related to Circles — Area of a Sector

17. **17.** The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

Answer:

$$\frac{154}{3} \text{ cm}^2$$

Solution:

1. Determine the angle swept by the minute hand in 60 minutes (360°), so in 1 minute it is 6° .
2. Calculate the angle for 5 minutes: $5 \times 6^\circ = 30^\circ$.
3. Apply the sector area formula with $r = 14$ cm and $\theta = 30^\circ$.

Derivation:

$$Area = \frac{30}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{1}{12} \times 22 \times 2 \times 14 = \frac{616}{12} = \frac{154}{3} \text{ cm}^2$$

Areas Related to Circles — Applications of Sector Area

18. **18.** Find the area of a quadrant of a circle whose circumference is 44 cm.

Answer:

$$38.5 \text{ cm}^2$$

Solution:

1. Use the circumference formula $2\pi r = 44$ to find the radius.
2. Solve for r : $r = \frac{44}{2\pi} = \frac{22}{\pi} = 7$ cm.
3. Calculate the area of the quadrant: $\frac{1}{4}\pi r^2$.

Derivation:

$$r = \frac{44 \times 7}{2 \times 22} = 7 \text{ cm. Area} = \frac{1}{4} \times \frac{22}{7} \times 7 \times 7 = \frac{154}{4} = 38.5 \text{ cm}^2$$

Areas Related to Circles — Area of a Quadrant

19. **19.** A chord of a circle of radius 10 cm subtends a right angle at the center. Find the area of the corresponding minor segment. (Use $\pi = 3.14$)

Answer:

$$28.5 \text{ cm}^2$$

Solution:

1. Find area of the sector: $\frac{90}{360} \times \pi r^2$.

2. Find area of the triangle formed by the chord and radii: $\frac{1}{2} \times r \times r$.
3. Subtract the triangle area from the sector area.

Derivation:

$$Area_{sector} = 0.25 \times 3.14 \times 100 = 78.5. \text{ Area}_{tri} = 0.5 \times 10 \times 10 = 50. \text{ Area}_{seg} = 78.5 - 50 = 28.5 \text{ cm}^2$$

Areas Related to Circles — Area of a Segment

- 20. 20.** The radii of two circles are 8 cm and 6 cm respectively. Find the radius of the circle having area equal to the sum of the areas of the two circles.

Answer:

$$10\text{cm}$$

Solution:

1. Set up the equation: $\pi R^2 = \pi r_1^2 + \pi r_2^2$.
2. Simplify to $R^2 = r_1^2 + r_2^2$.
3. Substitute values $r_1 = 8, r_2 = 6$ and solve for R .

Derivation:

$$R^2 = 8^2 + 6^2 = 64 + 36 = 100. \text{ Therefore, } R = \sqrt{100} = 10 \text{ cm.}$$

Areas Related to Circles — Areas of Circles

- 21. 21.** A chord AB of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

Answer:

$$Area \text{ of minor segment} = 20.4375 \text{ cm}^2; \text{ Area of major segment} = 686.0625 \text{ cm}^2$$

Solution:

1. Identify the radius $r = 15$ and central angle $\theta = 60^\circ$.
2. Calculate the area of sector $OAB = \frac{\theta}{360} \times \pi r^2$.
3. Calculate the area of $\triangle OAB$. Since $\theta = 60^\circ$ and $OA = OB$, it is an equilateral triangle, so $\text{Area} = \frac{\sqrt{3}}{4} r^2$.
4. Subtract the triangle area from the sector area to find the minor segment.
5. Subtract the minor segment area from the total circle area to find the major segment.

Derivation:

$$Area \text{ of sector} = \frac{60}{360} \times 3.14 \times 15^2 = \frac{1}{6} \times 3.14 \times 225 = 117.75 \text{ cm}^2. \text{ Area of } \triangle OAB = \frac{\sqrt{3}}{4} \times 15^2 = 0.432$$

Areas Related to Circles — Area of Segment

- 22. 22.** An umbrella has 8 ribs which are equally spaced. Assuming the umbrella to be a flat circle of radius 45 cm, find the area between the two consecutive ribs of the umbrella.

Answer:

$$Area = \frac{22275}{28} \text{ cm}^2 \text{ or } approx 795.54 \text{ cm}^2$$

Solution:

1. Determine the total number of sectors formed by 8 ribs, which is 8.
2. Calculate the central angle of each sector: $\theta = 360^\circ/8 = 45^\circ$.
3. Use the formula for the area of a sector: $A = \frac{\theta}{360} \times \pi r^2$.
4. Substitute $r = 45$ and $\theta = 45^\circ$ into the formula.
5. Simplify the expression to get the final area.

Derivation:

$$Area = \frac{45}{360} \times \frac{22}{7} \times 45 \times 45 = \frac{1}{8} \times \frac{22}{7} \times 2025 = \frac{11 \times 2025}{4 \times 7} = \frac{22275}{28} \text{ cm}^2.$$

Areas Related to Circles — Area of Sector

- 23. 23.** Find the area of the shaded region in a square $ABCD$ of side 14 cm, where four circles are drawn with centers at the four vertices of the square, each having a radius of 7 cm.

Answer:

$$Area = 42 \text{ cm}^2$$

Solution:

1. Calculate the area of the square $ABCD = \text{side}^2$.
2. Identify that each corner forms a quadrant of a circle of radius 7 cm.
3. Calculate the sum of the areas of the four quadrants, which is equivalent to the area of one full circle of radius 7 cm.
4. Subtract the area of the circle from the area of the square.
5. The remaining area is the shaded region.

Derivation:

$$Area \text{ of square} = 14 \times 14 = 196 \text{ cm}^2. \text{ Area of 4 quadrants} = 4 \times \left(\frac{1}{4} \pi r^2\right) = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

Areas Related to Circles — Combination of Plane Figures

- 24. 24.** A round table cover has six equal designs as shown in the figure. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of Rs. 0.35 per cm^2 . (Use $\sqrt{3} = 1.7$)

Answer:

$$Cost = \text{Rs. } 162.68$$

Solution:

1. Recognize that each of the 6 designs forms a segment of a circle.
2. Central angle for each segment is $360^\circ/6 = 60^\circ$.
3. Calculate the area of one segment: Area of sector - Area of equilateral triangle.
4. Multiply the area of one segment by 6 to get the total area of the designs.
5. Multiply the total area by the rate to find the cost.

Derivation:

$$\text{Area of one segment} = \frac{60}{360} \times \frac{22}{7} \times 28^2 - \frac{\sqrt{3}}{4} \times 28^2 = \frac{1}{6} \times \frac{22}{7} \times 784 - 1.7 \times 196 = 410.66 - 333.2 = 77.46 \text{ cm}^2$$

Areas Related to Circles — Area of Segment